

HW Chapter 2

1. FIFO: Turnaround = $\frac{0 + 200 + 400}{3} = 200$
 response = $\frac{0 + 200 + 400}{3} = 200$

SJF: Turnaround = $\frac{200 + 400 + 600}{3} = 400$
 response = $\frac{0 + 200 + 400}{3} = 200$

2. FIFO: Turnaround = $\frac{100 + 200 + 300}{3} = 200$
 response = $\frac{0 + 100 + 200}{3} = 100$

200 Turnaround = $\frac{200 + 400 + 600}{3} = 400$
 response = $\frac{0 + 200 + 400}{3} = 200$

300 Turnaround = $\frac{300 + 600 + 900}{3} = 600$
 response = $\frac{0 + 300 + 600}{3} = 300$

SJF: 1 Turnaround: $\frac{100+200+300}{3} = 200$

100

Response: $\frac{0+100+200}{3} = 100$

200 Turnaround: $\frac{200+400+600}{3} = 400$

Response: $\frac{0+200+400}{3} = 200$

300 Turnaround: $\frac{300+600+900}{3} = 600$

Response: $\frac{0+300+600}{3} = 300$

3. RR

Time slice: 1 100 Turnaround: $\frac{298+299+300}{3} = 299$

Response: $\frac{0+1+2}{3} = 1$

200 Turnaround: $\frac{598+599+600}{3} = 599$

Response: $\frac{0+1+2}{3} = 1$

300

Turnaround

$$\frac{843 + 219 + 907}{3}$$

3

= 849

$$\frac{849 - 300}{3} = 183$$

4. When the processes are added
at the same time SJF &
FIFO perform the same in
response time as well when
future processes added have long
run time than previous processes.

5. Very long quantum times will
be effectively makes it FIFO.

With this in mind we then
consider scheduling where all
processes added chronologically
also increase in time.

6. SJF with increasing times
functioning similarly to FIFO,
so the average response
time should in theory increase.

7. As quantum length increases,
the response time also increases.
As larger the quantum length, the
more the functions like $\phi(x)$.